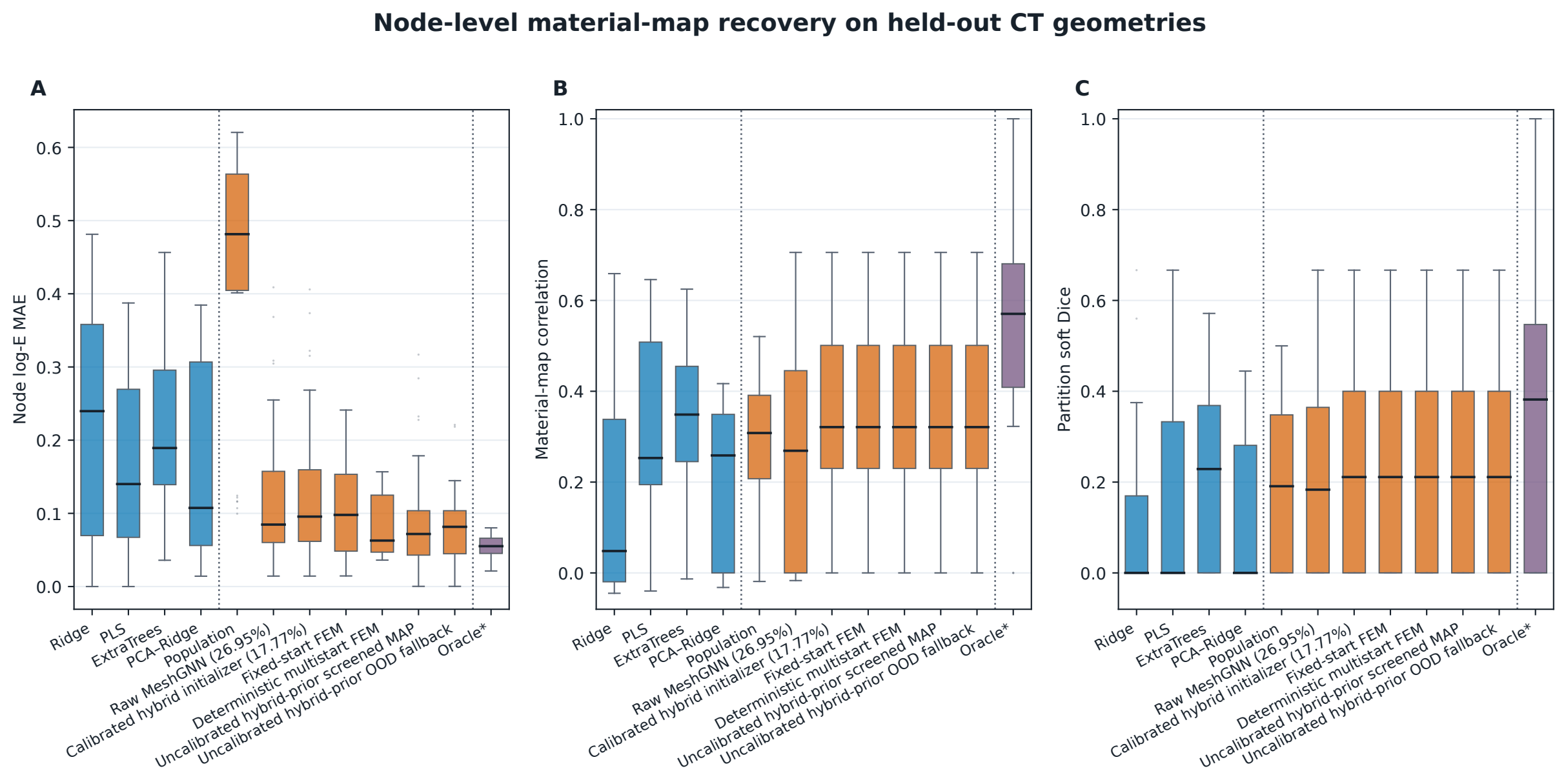
Node-level material-map recovery on held-out CT geometries
A
B
C
Node log-E MAE
Material-map correlation
Partition soft Dice
Ridge
PLS
ExtraTrees
PCA–Ridge
Population
Raw MeshGNN (26.95%)
Calibrated hybrid initializer (17.77%)
Fixed-start FEM
Deterministic multistart FEM
Uncalibrated hybrid-prior screened MAP
Uncalibrated hybrid-prior OOD fallback
Oracle*